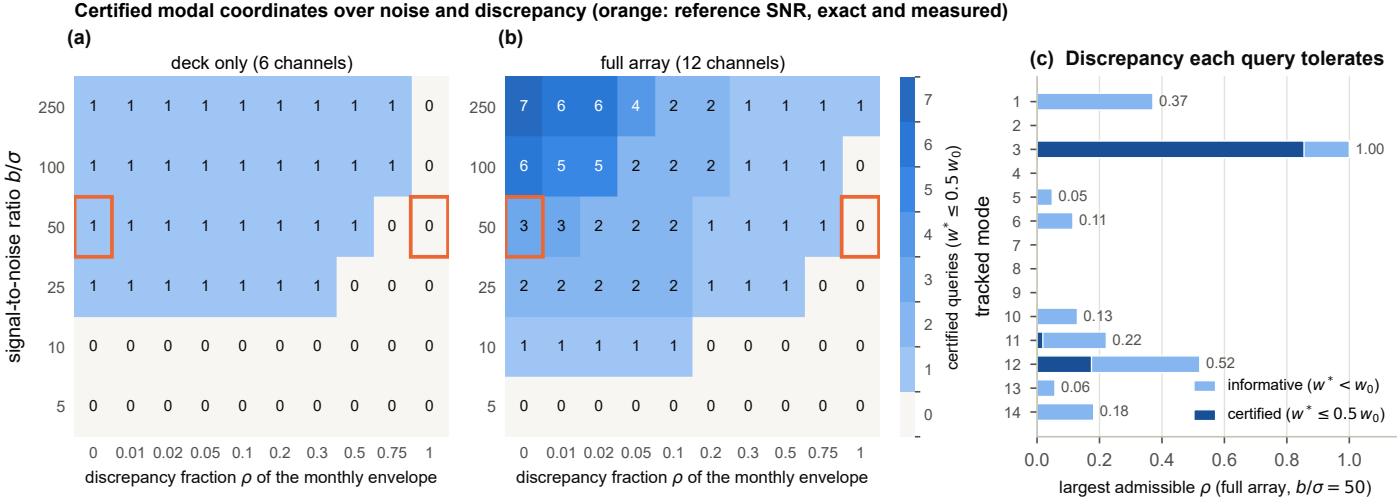

Certified modal coordinates over noise and discrepancy (orange: reference SNR, exact and measured)
(a)
deck only (6 channels)
(b)
full array (12 channels)
signal-to-noise ratio b/σ
250 1 1 1 1 1 1 1 1 1 0
100 1 1 1 1 1 1 1 1 1 0
50 1 1 1 1 1 1 1 1 0 0
25 1 1 1 1 1 1 1 0 0 0
10 0 0 0 0 0 0 0 0 0 0
5 0 0 0 0 0 0 0 0 0 0
250 7 6 6 4 2 2 1 1 1 1
100 6 5 5 2 2 2 1 1 1 0
50 3 3 2 2 2 1 1 1 1 0
25 2 2 2 2 2 1 1 1 0 0
10 1 1 1 1 1 0 0 0 0 0
5 0 0 0 0 0 0 0 0 0 0
0 0.01 0.02 0.05 0.1 0.2 0.3 0.5 0.75 1
discrepancy fraction ρ of the monthly envelope
0 0.01 0.02 0.05 0.1 0.2 0.3 0.5 0.75 1
discrepancy fraction ρ of the monthly envelope
certified queries ($w^* \le 0.5\, w_0$)
0 1 2 3 4 5 6 7
(c) Discrepancy each query tolerates
tracked mode
1 0.37
2
3 1.00
4
5 0.05
6 0.11
7
8
9
10 0.13
11 0.22
12 0.52
13 0.06
14 0.18
informative ($w^* < w_0$)
certified ($w^* \le 0.5\, w_0$)
0.0 0.2 0.4 0.6 0.8 1.0
largest admissible ρ (full array, $b/\sigma = 50$)